Bendrix Company — Overhead Cost Analysis

Excel Workbook Screenshots

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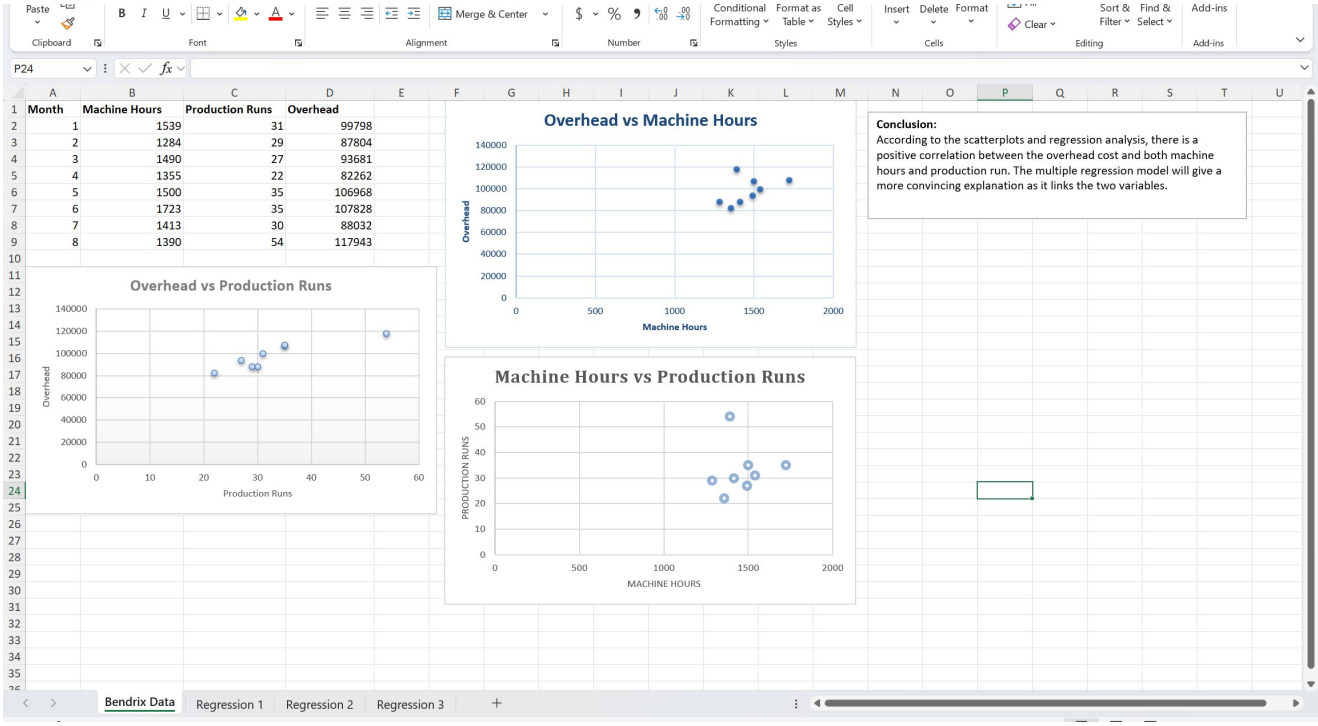
Bendrix Data & Scatterplots

Monthly data with three scatterplots visualising the relationships between Overhead, Machine Hours, and Production Runs.



Month	Machine Hours	Production Runs	Overhead
1	1539	31	99798
2	1284	29	87804
3	1490	27	93681
4	1355	22	82262
5	1500	35	106968
6	1723	35	107828
7	1413	30	88032
8	1390	54	117943

Regression 1 — Overhead vs Machine Hours

Simple linear regression with Machine Hours as the sole predictor. $R^2 = 0.2253$ (weak fit).

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.474656001
R Square	0.225298319
Adjusted R Square	0.096181372
Standard Error	11630.34748
Observations	8

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	236026129	236026129	1.744916716	0.23465904
Residual	6	811589895	135264982.5		
Total	7	1047616024			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	35022.31804	47882.73306	0.731418526	0.492089154	-82142.50895	152187.145	-82142.50895	152187.145
Machine Hours	43.11077952	32.63612022	1.320952957	0.23465904	-36.74692981	122.9684888	-36.74692981	122.9684888

Regression 2 — Overhead vs Production Runs

Simple linear regression with Production Runs as the sole predictor. $R^2 = 0.7850$ (strong fit).

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.885987991
R Square	0.784974719
Adjusted R Square	0.749137173
Standard Error	6127.314931
Observations	8

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	822352094.4	822352094.4	21.9036957	0.003395441
Residual	6	225263929.6	37543988.27		
Total	7	1047616024			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	60624.08643	8282.824943	7.319252411	0.000332136	40356.74392	80891.42895	40356.74392	80891.42895
Production Runs	1138.111439	243.1790121	4.680138427	0.003395441	543.0738326	1733.149046	543.0738326	1733.149046

Bendrix Data | Regression 1 | Regression 2 | Regression 3

Regression 3 — Multiple Regression

Multiple regression with both Machine Hours and Production Runs. $R^2 = 0.9428$ (excellent fit, both predictors significant).

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.970999617							
R Square	0.942840256							
Adjusted R Square	0.919976359							
Standard Error	3460.67807							
Observations	8							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	2	987734560.5	493867280.2	41.23707498	0.000781134			
Residual	5	59881463.53	11976292.71					
Total	7	1047616024						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	9163.872714	14616.86028	0.626938517	0.55820976	-28409.96282	46737.70825	-28409.96282	46737.70825
Machine Hours	36.23097671	9.749818048	3.716066959	0.01376807	11.16827153	61.29368188	11.16827153	61.29368188
Production Runs	1092.471394	137.8943793	7.922523022	0.000515905	738.0026076	1446.940181	738.0026076	1446.940181

Bendrix Data | Regression 1 | Regression 2 | Regression 3